3.1 Unit 1 BIOL1 Biology and disease

The digestive and gas exchange systems are examples of systems in which humans and other mammals exchange substances with their environment. Substances are transported from one part of the body to another by the blood system. An appreciation of the physiology of these systems requires candidates to understand basic principles including the role of enzymes as biological catalysts, and passive and active transport of substances across biological membranes.

The systems described in this unit, as well as others in the body, may be affected by disease. Some of these diseases, such as cholera and tuberculosis, may be caused by microorganisms. Other non-communicable diseases such as many of those affecting heart and lung function also have a significant impact on human health. Knowledge of basic physiology allows us not only to explain symptoms but also to interpret data relating to risk factors.

The blood has a number of defensive functions which, together with drugs such as antibiotics, help to limit the spread and effects of disease. It is anticipated that the smaller size of this unit will allow opportunity for the development of the skills of application and analysis as well as for the acquisition of the investigatory skills associated with *Investigative and practical skills* detailed in Unit 3.